

FORESIGHT

Exploratory Data Analysis Report

AI-Powered Demand & Inventory Intelligence Platform

Purpose

This report defines the data-quality and exploratory checks that prepare supermarket sales records for SKU-level demand forecasting. The analysis is designed to support the FORESIGHT workflow: clean sales data, identify demand patterns, train a model, predict demand, score inventory risk and recommend action.

Field	Role in analysis
date	Build daily demand series and calendar features
sku / product	Forecast each stock keeping unit separately
quantity_sold	Forecast target
current_stock	Calculate days of cover and reorder exposure
holiday / season / weather	Optional demand drivers

Analysis checks

1. Data quality

Validate mandatory columns, parse dates, remove duplicate SKU-day records, and exclude negative demand or stock values.

2. Demand behaviour

Review trend, weekly seasonality, holiday uplift and SKU concentration. Use product-level plots to investigate unusually high or low sales.

3. Feature readiness

Create day-of-week, month, weekend, 7-day lag and rolling-demand features. Keep sufficient historical records before model training.

4. Forecast evaluation

Use a chronological holdout set and MAPE to evaluate demand forecasts. Compare model accuracy to a rolling-average baseline.

5. Inventory decision rules

Use forecasted daily demand to calculate days of cover. Flag stockout risk below supplier lead time and overstock above 45 days of cover.

Recommended next step

Load the cleaned data in the Streamlit dashboard, review forecast accuracy and address the highest-priority reorder recommendations before the next supplier order cycle.